

Example 2: Fetch from Statistics Estonia with R

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```
library(quarto)      # for compiling Quarto presentations
library(pxweb)
library(tidyverse)   # dplyr, ggplot2, readr, etc.
library(here)        # path
```

Accessing Statistics Estonia Data with pxweb

- ▶ Statistics Estonia is **not yet part of DBnomics**
- ▶ No dedicated R package for andmed.stat.ee
- ▶ But: you can access data using the pxweb R package
- ▶ Based on the **PX-WEB API**, which is also used by many Nordic stats agencies

How to Identify Variables

How to Identify Query Variables

- Use either:
1. Programmatically via the metadata (`pxweb_get()`, `pxweb_get_metadata()`)
 2. From the Statistics Estonia web interface – click **“API query for this table”**

```
pa111_meta <- pxweb_get(  
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111")  
pa111_meta
```

PXWEB METADATA

PA111: AVERAGE MONTHLY GROSS WAGES (SALARIES), MEDIAN, DEC
variables:

[[1]] Näitaja: Indicator

[[2]] Tegevusala: Economic activity

[[3]] Vaatlusperiood: Reference period

```
pa111_meta$variables[[1]]
```

\$code

```

px_query_list1 <- list(
  "Näitaja" = c("GR_W_D1", "GR_W_D5", "GR_W_D9"),
  "Tegevusala" = "*",
  "Vaatlusperiood" = "*"
)

pa111 <- pxweb_get(
  url = "https://andmed.stat.ee/api/v1/en/stat/PA111",
  query = px_query_list1
)

# Convert to data frame
pa111.df <- as.data.frame(pa111, column.name.type = "text")
pa111.code <- as.data.frame(pa111, column.name.type = "code")
pa111.df$Näitaja <- pa111.code$Näitaja

selected_sectors <- c("Manufacturing", "Construction", "Int
# Clean and reshape
pa111_plotdata <- pa111.df %>%
  mutate(

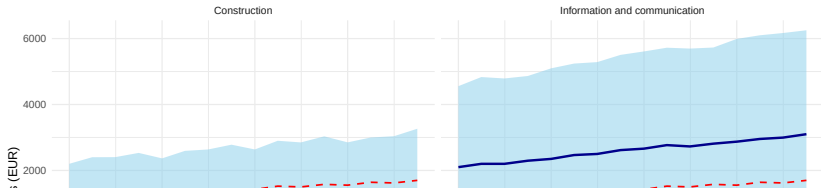
```

```
overall_median <- pa111_plotdata %>%  
  filter(`Economic activity` == "Total - all activities") %  
  select(quarter, total_median = Median)
```

```
pa111_plotdata <- pa111_plotdata %>%  
  filter(`Economic activity` != "Total - all activities") %  
  left_join(overall_median, by = "quarter")
```

```
ggplot(pa111_plotdata, aes(x = quarter)) +
  geom_ribbon(aes(ymin = D1, ymax = D9), fill = "skyblue",
  geom_line(aes(y = Median), color = "darkblue", linewidth
  geom_line(aes(y = total_median), color = "red", linetype
  facet_wrap(~`Economic activity`, ncol = 2) +
  labs(
    title = "Wage Distribution by Sector (1st to 9th Decile
    subtitle = "Red dashed line shows overall median (Total
    x = "Quarter",
    y = "Gross Nominal Monthly Wages (EUR)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Wage Distribution by Sector (1st to 9th Decile, Median)
Red dashed line shows overall median (Total – all activities)



Rendering All Formats from R or Bash

To render the document to **all output formats** (docx, Beamer, Reveal.js): run programmatically from another script or the R console:

```
quarto::quarto_render(here("teaching_materials", "example2_
```

or from the command line:

```
quarto render example2_statistics_estonia.qmd
```